Third Assginment

16.10.2024

Sent **8/8 Points**

Forsøk 1

Se gjennom tilbakemelding
17.10.2024

Forsøk 1 Poengsum:
8/8

Vis tilbakemelding

Anonym vurdering: **nei**

3 tillatte forsøk
8.10.2024 til 17.10.2024

Detaljer

Assignment Title: Implementing and Analyzing the Quick Sort with Random Pivot (**Randomized Quick Sort**) Algorithm in C++

Assignment Description
In this assignment, you will implement and analyze the **Randomized Quick Sort** algorithm in C++. Randomized Quick Sort is a variation of the classic Quick Sort algorithm that uses a randomly selected pivot to enhance the average-case performance and avoid worst-case scenarios on certain inputs.

Implementation
Write a C++ program that implements the Randomized Quick Sort algorithm. Your program should include the following components:

1. A function to perform the **Randomized Quick Sort** to sort an unsorted array.
2. Input and output mechanisms to take user input for an unsorted array and display the sorted array.
3. Ensure that your program handles integers and floating-point numbers based on user input.
4. Implement error handling to manage edge cases gracefully (e.g., when the array is empty or input is invalid).

Testing and Analysis

1. The program should ask the user to input an unsorted array.
2. Call the Randomized Quick Sort function to sort the array and output the sorted result.
3. Perform a complexity analysis of the Randomized Quick Sort algorithm (Mathematical Prove), and compare its performance (execution time) with the following algorithms:
   - Merge Sort
   - Normal Quick Sort

Testing Example

- **Input**:
  - Unsorted Array: [29, 10, 14, 37, 13, 7, 1, 20]
- **Output**:
  - The sorted array is: [1, 7, 10, 13, 14, 20, 29, 37]
- **Explanation**:
  - The array is sorted in ascending order using the Randomized Quick Sort algorithm, which randomly selects a pivot at each recursive step to avoid worst-case time complexity.

Documentation and Submission
Include comments and documentation within your code to explain the logic and purpose of each component.

Write a brief report that includes:

1. A description of the Randomized Quick Sort algorithm and how it works.
2. Results of the performance analysis, is it possible to optimize this algorithm? and how?
3. Discuss the efficiency of Randomized Quick Sort in comparison to Merge Sort and Normal Quick Sort based on your results.

Submit your C++ code files and the report as a single compressed file (ZIP file) through your Canvas group.

Grading Criteria
Your assignment will be evaluated based on the following criteria:

1. Correct implementation of the Randomized Quick Sort algorithm that handles handle integers and floating-points (3 Points).
2. Effective error handling (1 Point).
3. Clarity and readability of the code, including comments and documentation (1 Point).
4. Accuracy and clarity of the performance analysis report (worst time complexity discussion included) in a PDF file (3 Points).